CSA07 – Computer Networks

Annexure A: Concept Mapping Assignment

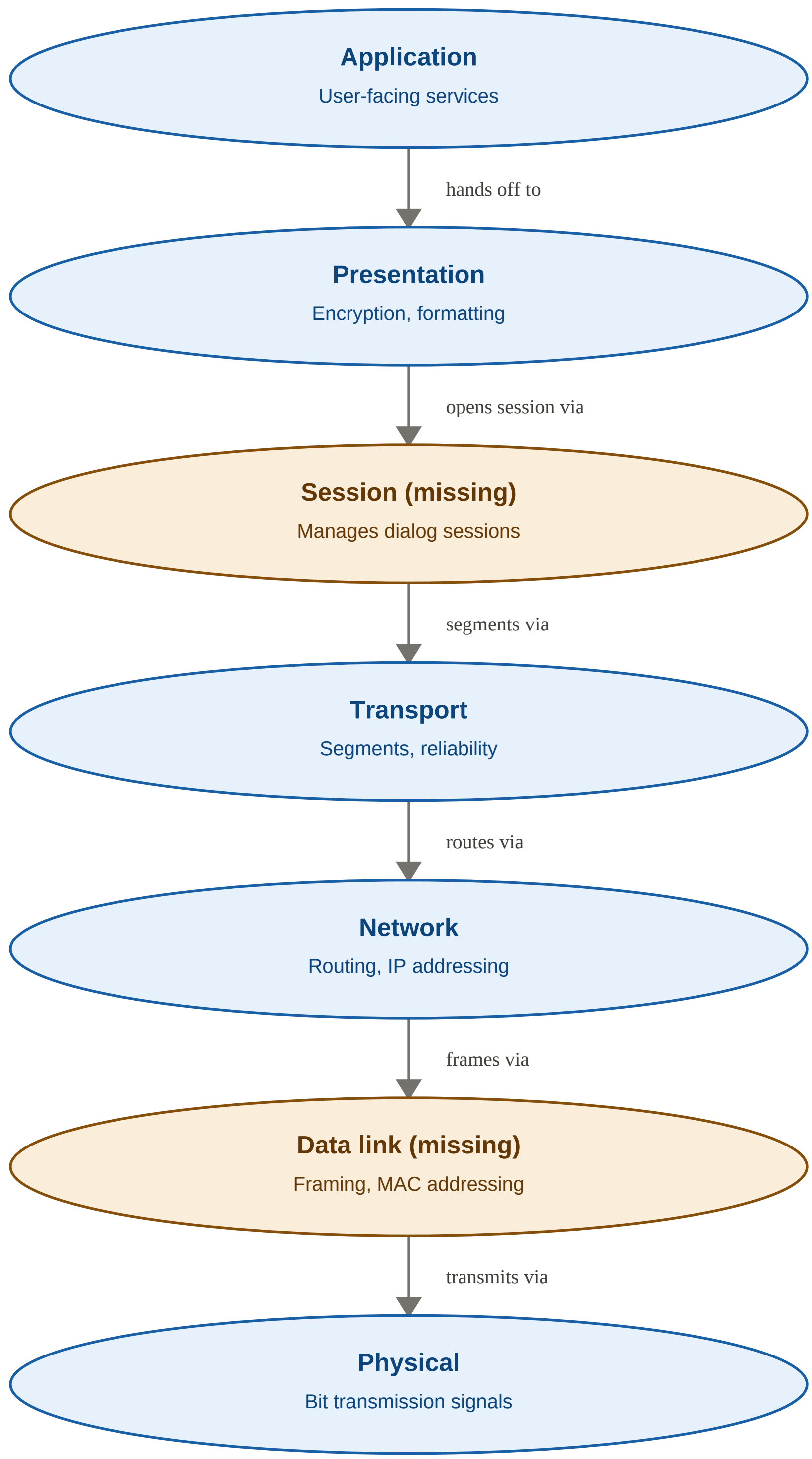
CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Code of Conduct

I KAVINILAVU S(192565038) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

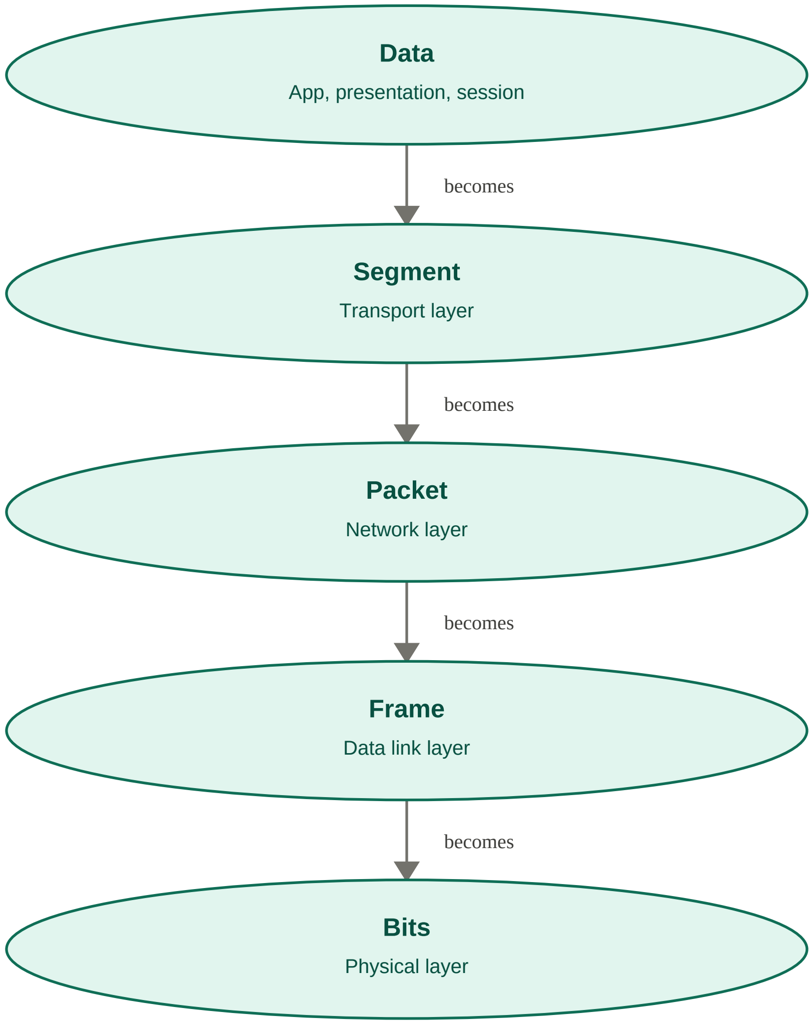
Signature of the student: ____________________

1. Completing the concept map (missing layers)



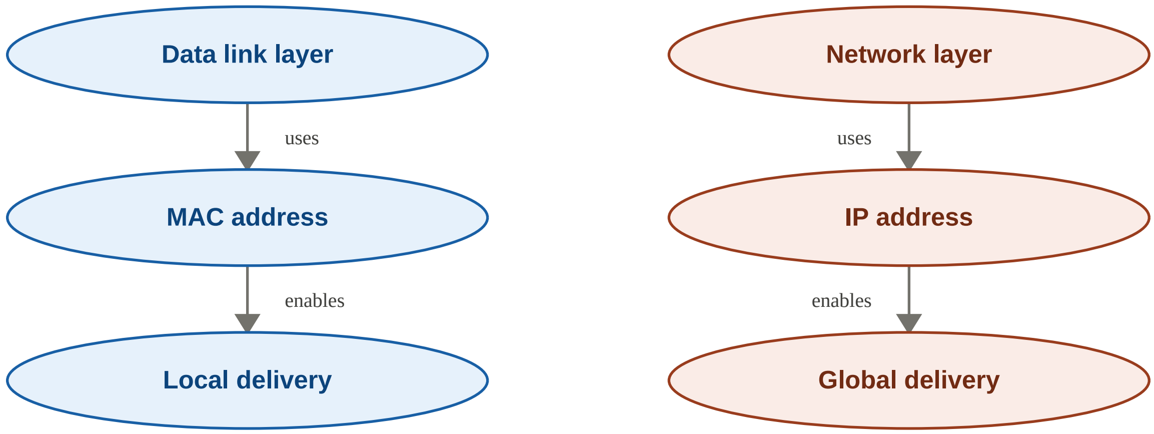
The two missing layers are the **Session layer** and the **Data Link layer**. Session sits between Presentation and Transport: it establishes, manages, and terminates the dialog between the two application processes, and provides synchronization checkpoints so a broken exchange can resume rather than restart. Data Link sits between Network and Physical: it frames packets with source and destination MAC addresses, controls access to the shared medium, and performs hop-by-hop error detection. Without Session there is no defined way to open, coordinate, and recover a conversation between programs; without Data Link, bits from the Physical layer have no reliable way to be checked for corruption or delivered to the correct device on the local network. Both are essential for a complete, reliable path from application to medium.

2. Concept map: data units vs. OSI layers



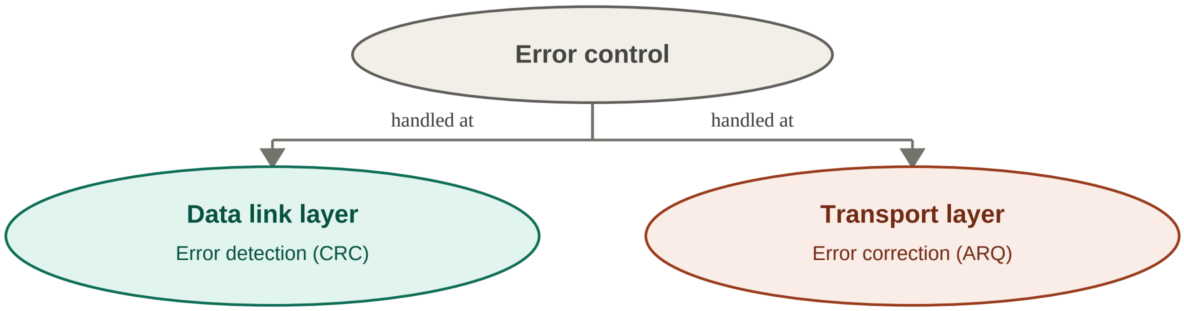
As data descends the stack, each layer **encapsulates** the unit handed down from above by wrapping it with its own header (and sometimes a trailer). Transport wraps the message into a Segment, adding port numbers and control information for process-to-process delivery. Network wraps the Segment into a Packet, adding IP addresses for host-to-host delivery. Data Link wraps the Packet into a Frame, adding MAC addresses and an error-check trailer for node-to-node delivery. Physical converts the Frame into Bits – actual signals for the medium. At the receiving end, the reverse process (de-encapsulation) strips each header in order, letting one continuous piece of application data be correctly addressed, routed, checked, and transmitted.

3. Local delivery (MAC) vs. global delivery (IP)



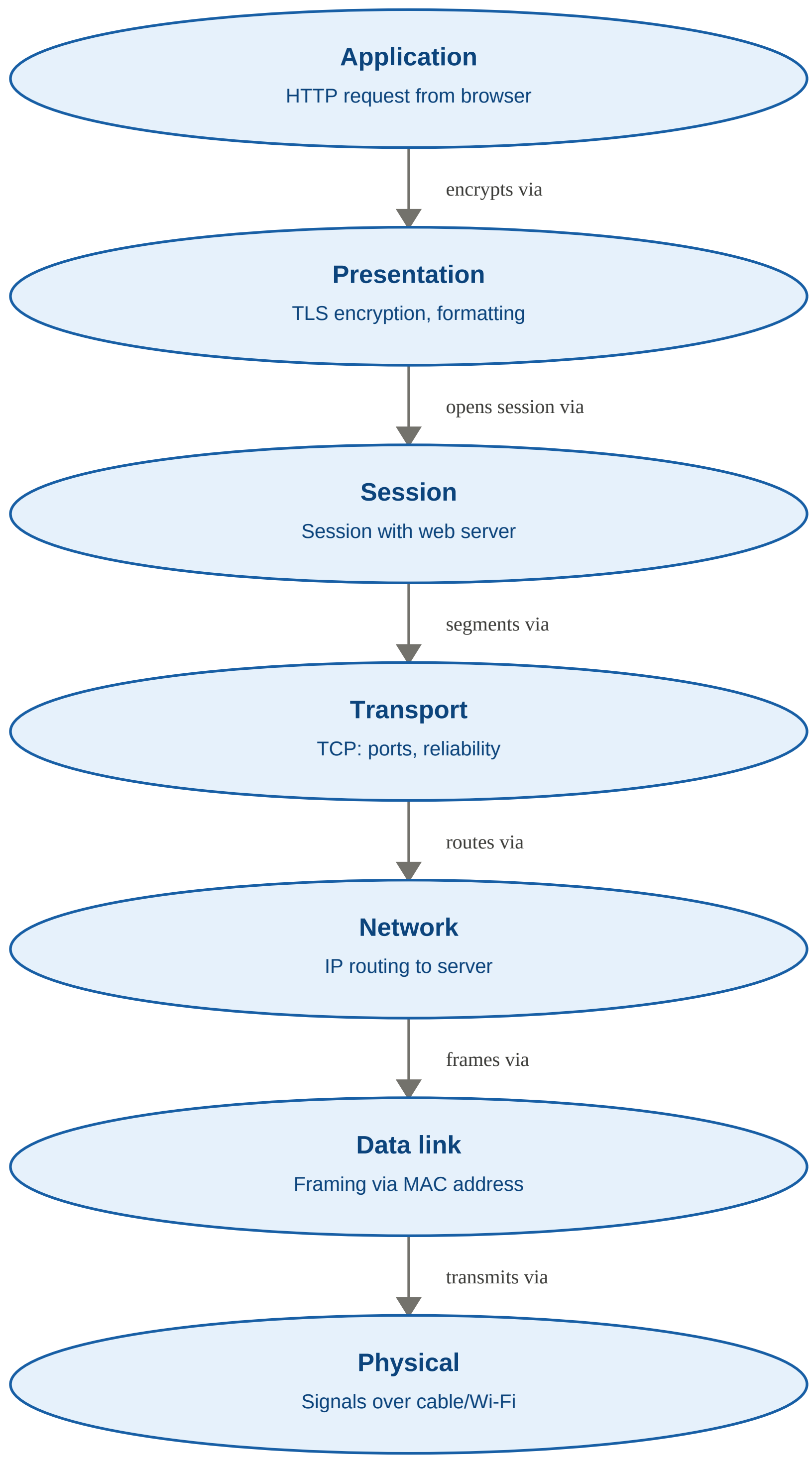
The **Data Link layer** uses **MAC addresses** – physical, hardware addresses that are only meaningful within a single local network segment – to achieve **local delivery** to the next adjacent device. The **Network layer** uses **IP addresses** – logical addresses that stay valid across the entire path – to achieve **global delivery** between hosts on different networks. A packet's IP address never changes for the whole journey, but its MAC address changes at every hop as routers re-frame it for the next local link (resolved via ARP). In effect, global delivery is achieved by chaining together a series of local deliveries: IP addressing decides the overall destination, MAC addressing decides who gets it next.

4. Error detection (Data Link) vs. error correction (Transport)



The **Data Link layer** performs **error detection** on each individual link using CRC or checksums – a hop-by-hop check that simply discards a corrupted frame. The **Transport layer** (e.g. TCP) performs **error correction** end-to-end using sequence numbers, acknowledgments, and retransmission (ARQ), recovering lost or corrupted segments across the entire path regardless of how many links lie in between. Not every link is guaranteed reliable, so Data Link detection alone cannot guarantee delivery. Together they form a layered defense: errors are caught early and cheaply at the link, while Transport still guarantees the application receives complete, correct data end-to-end.

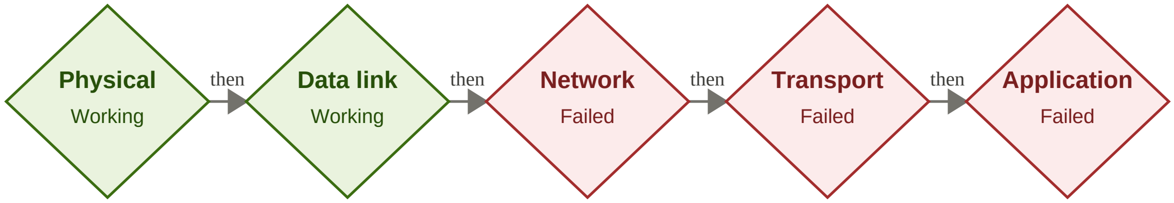
5. Concept map for web browsing using OSI layers



Every layer here depends completely on the one below it. If **Physical** fails (cable unplugged, no signal), nothing moves at all. If **Data Link** fails, frames never reach the local router. If **Network** fails, the request is misrouted or lost. If **Transport** fails, the connection times out or is refused even though the host is reachable. If

Session/Presentation fails (a TLS handshake breaking), the browser blocks the page with a certificate warning even though every lower layer worked. If **Application** fails, the server itself returns an error such as 404 or 500. A single broken layer breaks the entire page load, even when every other layer is functioning correctly.

6. Troubleshooting with the concept map



Green = layer confirmed working, red = layer failing

Physical and Data Link are confirmed working, so the local link is healthy – the cable or Wi-Fi signal is present and frames are delivered correctly within the local network. The fault first appears at the **Network layer**, pointing to a routing or IP configuration problem: a wrong subnet mask, a missing default gateway, a router failing to forward packets, or an unreachable destination network. Transport and Application are not independent failures – they fail only because packets never got past the Network layer. This is exactly why the concept map helps troubleshooting: check layers bottom-up, stop at the first one that fails, and you know immediately where to look – there is no need to inspect browser or application settings when the real fault is in IP routing.